NEWTON SCHOOL

MATH | ANSWER KEY

Area of Shapes — Answer Key

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Part A — Rectangles & Squares

Problem 1 — Rectangle 5 m × 6 m

Multiply numbers AND units

Visual Count the 1 m^2 unit squares inside: 5 rows × 6 cols.

Math $5\,m \times 6\,m = 30\,m^2$ (number: 5·6=30; unit: m·m = m^2)

30 m^2

Problem 2 — Rectangle 16 yd × 9 yd

Rectangle area

Math $16 \times 9 = 144;\ \ yd \times yd = yd^2$

144 yd^2

Problem 3 — Rectangle 3.5 cm × 8 cm

Decimal × integer

Math $3.5 \times 8 = 28$; unit cm^2

28 cm^2

Problem 4 — Square side 12 in

$s \times s = s^2$

Math $12 \times 12 = 144$; unit in^2

144 in^2

Problem 5 — Square side 5⅙ mm

Mixed-number square

Convert $5\frac{1}{6} = \dfrac{31}{6}$

Square $\left(\dfrac{31}{6}\right)^2 = \dfrac{961}{36}$

Mixed $961 \div 36 = 26\,\text{remainder}\,25 \to 26\frac{25}{36}$

26 25/36 mm²

Part B — Triangles

Problem 6 — Triangle b=10 km, h=4 km

Half of a rectangle

Visual The triangle fills half of a 10×4 rectangle.

Math $\dfrac{10 \times 4}{2} = \dfrac{40}{2} = 20$; unit km²

20 km²

Problem 7 — Triangle b=9 in, h=5.2 in

Use perpendicular height (NOT slant)

Trap The 6 in is the slanted side — ignore it.

Math $\dfrac{9 \times 5.2}{2} = \dfrac{46.8}{2} = 23.4$; unit in²

23.4 in²

Problem 8 — Triangle b=11.6 km, h=6 km

Half of rectangle

Math $\dfrac{11.6 \times 6}{2} = \dfrac{69.6}{2} = 34.8$; unit km²

34.8 km²

Problem 9 — Triangle b=8 yd, h=5 yd

Math $\dfrac{8 \times 5}{2} = \dfrac{40}{2} = 20$; unit yd²

20 yd²

Part C — Parallelograms

Problem 10 — Parallelogram b=9.7 yd, h=5 yd

Parallelogram = rectangle in disguise

Visual Slice the slanted end, slide it across → 9.7 × 5 rectangle.

Math $9.7 \times 5 = 48.5$; unit yd² (the 5.8 yd slant is a distractor)

48.5 yd²

Problem 11 — Parallelogram b=12 km, h=8 km

Math $12 \times 8 = 96$; unit km²

96 km²

Problem 12 — Parallelogram b=8 m, h=6 m

Math $8 \times 6 = 48$; unit m²

48 m²

Problem 13 — Parallelogram b=14 cm, h=9 cm

Math $14 \times 9 = 126$; unit cm²

126 cm²

Part D — Trapezoids

Problem 14 — Trapezoid b_1=3, b_2=6, h=4 (cm)

Two trapezoids → parallelogram

Visual Two of these make a 9×4 parallelogram = 36 cm². One trapezoid = half.

Math $\dfrac{(3+6)\times 4}{2} = \dfrac{36}{2} = 18$; unit cm²

18 cm²

Problem 15 — Trapezoid b_1=7, b_2=4, h=5 (mm)

Math $\dfrac{(7+4)\times 5}{2} = \dfrac{55}{2} = 27.5$; unit mm²

27.5 mm²

Problem 16 — Trapezoid $b_1=6$, $b_2=3$, $h=3.9$ (m)

Math $\frac{(6+3)\times 3.9}{2} = \frac{9\times 3.9}{2} = \frac{35.1}{2} = 17.55$; unit m²

17.55 m²

Problem 17 — Trapezoid $b_1=4$, $b_2=1$, $h=3.2$ (yd)

Math $\frac{(4+1)\times 3.2}{2} = \frac{5\times 3.2}{2} = \frac{16}{2} = 8$; unit yd²

8 yd²

Problem 18 — Trapezoid $b_1=7$, $b_2=3\frac{1}{3}$, $h=4\frac{2}{3}$ (mm)

Mixed-number trapezoid

Convert $b_2 = \frac{10}{3},\ h = \frac{14}{3},\ b_1 = 7 = \frac{21}{3}$

Sum $b_1 + b_2 = \frac{21}{3} + \frac{10}{3} = \frac{31}{3}$

Times h $\frac{31}{3} \times \frac{14}{3} = \frac{434}{9}$

Half $\frac{434}{9} \div 2 = \frac{434}{18} = \frac{217}{9} = 24\frac{1}{9}$

24 1/9 mm²

Part E — Word Problems

Problem 19 — Trapezoid: area = 96 cm², $h = 6$ cm, $b_2 = b_1 + 4$. Find b_1, b_2.

Reverse the formula

Setup $\frac{(b_1+b_2)\times 6}{2} = 96 \Rightarrow (b_1+b_2)\times 3 = 96$

Solve $b_1 + b_2 = 32$

Substitute $b_1 + (b_1 + 4) = 32 \Rightarrow 2b_1 = 28 \Rightarrow b_1 = 14$

Therefore $b_2 = 14 + 4 = 18$

b_1 = 14 cm, b_2 = 18 cm

Problem 20 — Parallelogram floor 14 m × 5 m

Parallelogram = rectangle

Math $14 \times 5 = 70$; unit m²

70 m² of carpet